

NOIDA INSTITUTE OF ENGINEERING AND TECHNOLOGY
GREATER NOIDA-201306
(An Autonomous Institute)
School of Electronic Engineering

Department of EE (VLSI)

Session (2025-2026)
Advance Engineering Mathematics
Project Based Assessment Using Python

(BAS0406)

(4th Semester)

Submitted To:
Dr. Aakansha Vyas

Submitted By:
Name: Aditya Raj
Roll No: 2401331730005



NIET
Greater Noida
Autonomous Institute



*Affiliated to Dr. A. P. J Abdul Kalam Technical University, Uttar Pradesh,
Lucknow*

NOIDA INSTITUTE OF ENGINEERING AND TECHNOLOGY
GREATER NOIDA-201306
(An Autonomous Institute)

DECLARATION

I hereby declare that the project entitled “**Survey on Music Listening Habits Using Statistical Analysis and Python**” is an authentic record of my own work carried out by me during the academic session 2025–2026 as part of the course requirements for Advance Engineering Mathematics (BAS0406). The work has been completed under the guidance and supervision of my faculty, and all the efforts put into this project are original and genuine.

I further declare that the data used in this project has been collected through a Google Form survey, and the analysis has been med using Python programming and appropriate statistical methods such as mean, variance, standard deviation, skewness, and kurtosis. The results presented in this report are based on my own analysis and interpretation of the collected data.

I confirm that this project report has not been submitted in part or full to any other university or institution for the award of any degree or diploma. All sources of information used in this project have been properly acknowledged wherever required.

I take full responsibility for the authenticity and accuracy of the work presented in this report.

Name: Aditya Raj

Roll No.: 2401331730005

ACKNOWLEDGMENT

I would like to express my sincere gratitude to **Dr. Akansha Vyas** for her invaluable guidance, constant supervision, and continuous encouragement throughout the completion of this project titled *“Project-Based Assessment Using Python.”* Her insightful suggestions, constructive feedback, and academic support played a crucial role in shaping the direction and quality of this study.

I am also deeply thankful to **Dr. Rajan Singh**, Head of the Department, for his motivation, encouragement, and institutional support. His guidance and positive outlook greatly contributed to the successful completion of this project.

I would also like to extend my appreciation to the **Noida Institute of Engineering & Technology, Greater Noida**, for providing the academic environment and resources necessary to carry out this study effectively.

Finally, I am grateful to all the respondents who participated in the Google Form survey and provided valuable responses that made this statistical analysis possible. Their cooperation and honest feedback were essential for completing this project successfully.

This project has provided me with a valuable opportunity to apply statistical concepts and Python programming to real-world data analysis, thereby enhancing my understanding of statistical interpretation and research methodology

INDEX

S.No	Topic	Page No	Signature
1.	Introduction	3	
2.	Survey on Music Listening Habits Using Statistical Analysis and Python • Objective • Methodology • Interpretation • Expected Outcomes	3-5	
3.	Google Form Question(Screenshot)	6-7	
4.	Pie Chat of Response(Screenshot)	8-9	
5.	Spreadsheets of Response	10	
6.	Python Code	11	
7.	Solution of Questions	12-20	
8.	Python code Relation Between Question	21	
9.	Relation Between Question	22-23	
10.	Conclusion	24	

INTRODUCTION

Music plays an important role in people's daily lives. It helps individuals relax, reduce stress, improve mood, and increase concentration while **studying** or **working**. With the growth of digital platforms such as **Spotify**, **YouTube Music**, and **JioSaavn**, people now have easy access to a wide variety of music anytime and anywhere.

Understanding people's music listening habits can provide useful insights about their preferences, **listening time**, **favorite genres**, and reasons for listening to music. Surveys are an effective method to collect such data from different individuals.

In this project, data is collected through an online Google Form survey related to **music listening habits**. The responses are analyzed using statistical methods such as mean, median, mode, variance, standard deviation, quartile deviation, skewness, and kurtosis.

Python programming is used to perform statistical calculations and generate graphical representations like pie charts.

The purpose of this project is to analyze people's music listening behavior using statistical analysis and programming tools

Survey on Music Listening Habits Using Statistical Analysis and Python

Objective:

The objective of this study is to analyze people's music listening habits using statistical methods and Python programming.

Specific Objectives

1. To identify the most preferred music genres among respondents.
2. To analyze how frequently people listen to music.
3. To determine the most common situations where people listen to music.
4. To examine how much people spend on music subscriptions.
5. To identify whether people prefer listening to music alone or with others.
6. To analyze the main reasons why people listen to music.
7. To determine the average daily time spent listening to music.

Methodology:

1. Data Collection:

1. The data for this study was collected through an **online Google Form survey** related to music listening habits.
2. The questionnaire included multiple-choice questions about:
 3. Favorite music genre
 4. Frequency of listening
 5. Preferred listening location
 6. Monthly spending on music apps
 7. Preferred listening company
 8. Reason for listening to music
 9. Daily listening duration

Around **54 responses** were collected from participants.

2. Data Organization:

The data was collected using a **Google Form survey** about music listening habits. The responses were exported into an Excel spreadsheet. Each question was arranged in separate columns such as music genre preference, listening frequency, listening place, spending on music apps, listening companion, reason for listening to music, and daily listening duration. This organized data helped in performing statistical analysis using Python.

3. Statistical Calculations:

• **Mean ($\bar{X}$)** = $\sum f x / \sum f$

• **Median (M)** = $L + [(N/2 - cf) / f] \times h$

- **Mode (m)** = $L + [(f_1 - f_0) / (2f_1 - f_0 - f_2)] \times h$

- **Variance (σ^2)** = $\Sigma(x - \bar{x})^2 / n$

- **Standard Deviation (σ)** = $\sqrt{[\Sigma(x - \bar{x})^2 / n]}$

- **Quartile Deviation(Q.D.)** = $Q_3 - Q_1 / 2$

- **Skewness (β_1)** = $(\mu_3)^2 / (\mu_2)^3$

- **Kurtosis (β_2)** = $\mu_4 / (\mu_2)^2$

Interpretation:

The statistical results were interpreted to understand patterns in music listening habits.

From the analysis, it was observed that:

- Most respondents listen to music daily.
- Bollywood and pop are the most preferred music genres.
- Many people listen to music for relaxation and stress relief.
- Most respondents listen to music using mobile phones.
- The majority of people listen to music for about 1–3 hours daily..

Expected Outcomes

The study will:

- To identify the most preferred music genres among people.
- To understand how frequently people listen to music.
- To analyze the average time people spend listening to music daily.
- To understand the main reasons why people listen to music

Google Form

Survey on Music Listening Habits Using Statistical Analysis and Python (Question Screenshots)

Music Listening Habits Survey

This survey is created to understand people's music listening habits and preferences in daily life

Name *

Short answer text

Age Group *

- ☐ Under 18
- ☐ 18-20
- ☐ 20-25
- ☐ Above 25

Do you listen to music daily? *

- ☐ Yes
- ☐ No
- ☐ Sometimes

How many hours do you listen to music daily? *

- ☐ Less than 1 hour
- ☐ 1-2 hrs
- ☐ 2-3 hrs
- ☐ More than 3 hrs

Which type of music do you like the most? *

- ☐ Bollywood
- ☐ Pop
- ☐ Rap / Hip-Hop
- ☐ Other

When do you usually listen to music? *

- ☐ While studying
- ☐ While traveling
- ☐ While exercising
- ☐ Before sleeping

Which device do you use to listen to music *

- ☐ Mobile phone
- ☐ Laptop
- ☐ Smart speaker
- ☐ Other

Which music app do you use the most? *

- ☐ Spotify
- ☐ YouTube Music
- ☐ Apple Music
- ☐ Other

Does music help you reduce stress? *

- ☐ Yes
- ☐ No
- ☐ Sometimes

Who is your favorite singer or band?



Short answer

Short answer text



Required

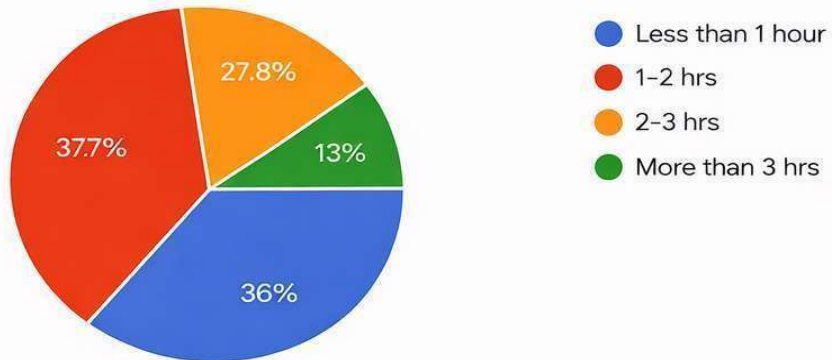


Pie Chart Of Responses(ScreenShot)

1. How many hours do you listen to music daily?

 [Copy chart](#)

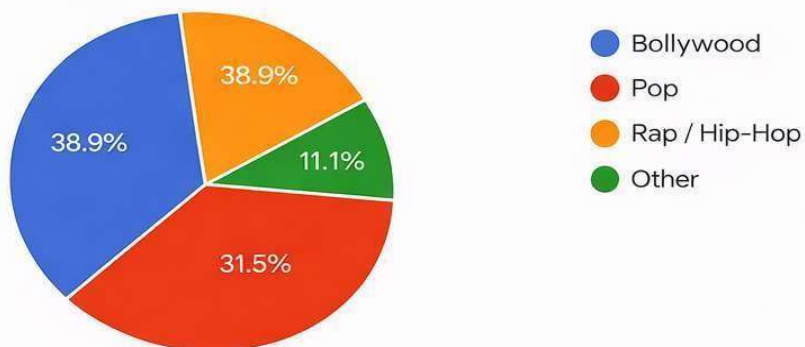
54 responses



2. Which type of music do you like the most?

 [Copy chart](#)

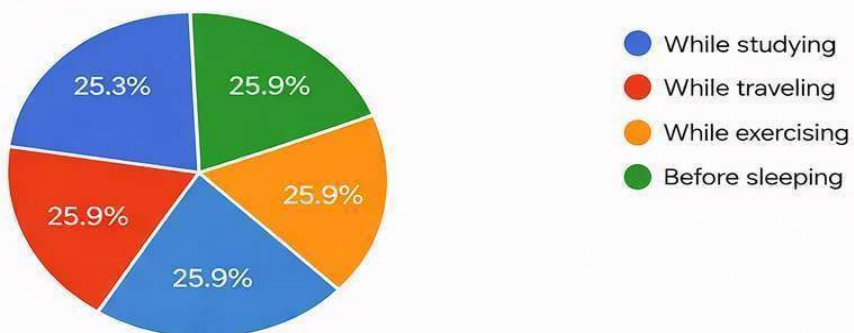
54 responses



3. When do you usually listen to music?

 [Copy chart](#)

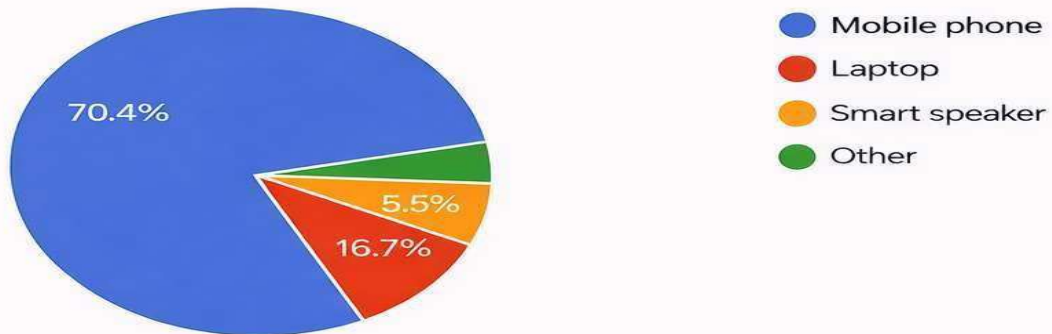
54 responses



1. Which device do you use to listen to music?

 [Copy chart](#)

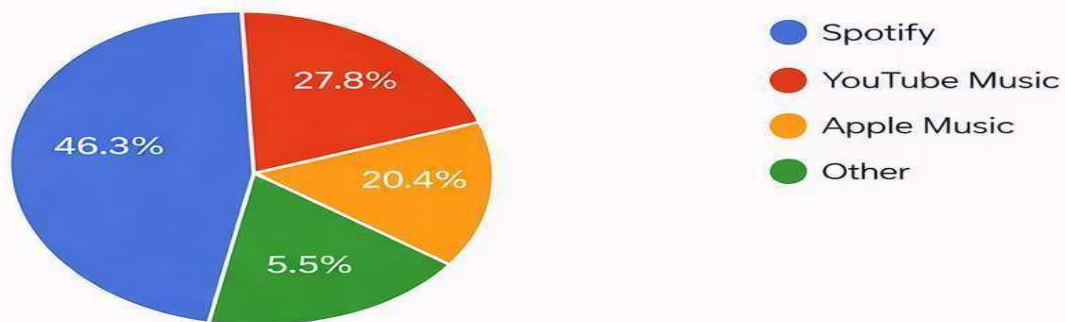
54 responses



2. Which music app do you use the most?

 [Copy chart](#)

54 responses



Spreadsheet of Response

Timestamp											
A	B	C	D	E	F	G	H	I	J	K	L
1	Timestamp	Name	Age Group	How many hrs	Which type	When do	Which device	Which music	Does music help you reduce stress?		
2	03/03/20	Harsh V	18-20	1-2 hrs	Rap / Hip	Before s	Laptop	Apple M	No		
3	03/03/20	Deepak V	Under 18	1-2 hrs	Other	While tra	Laptop	Spotify	Sometimes		
4	02/03/20	Lakshay	Under 18	1-2 hrs	Rap / Hip	While stu	Mobile p	YouTube	Sometimes		
5	04/03/20	Agush S	18-20	2-3 hrs	Other	Before s	Mobile p	Other	No		
6	01/03/20	Ankit M	Above 21	1-2 hrs	Rap / Hip	While stu	Smart sp	Spotify	Sometimes		
7	02/03/20	Raghav V	20-25	1-2 hrs	Other	Before s	Laptop	Spotify	Sometimes		
8	03/03/20	Rajat Ch	Above 21	Less than	Rap / Hip	Before s	Laptop	Apple M	No		
9	04/03/20	Vedant N	Above 21	1-2 hrs	Pop	While tra	Mobile p	Apple M	Yes		
10	03/03/20	Rudra S	20-25	1-2 hrs	Other	While tra	Laptop	Other	Yes		
11	03/03/20	Pigush V	18-20	More than	Pop	While exe	Mobile p	YouTube	No		
12	02/03/20	Nikhil T	Above 21	More than	Rap / Hip	Before s	Laptop	YouTube	Sometimes		
13	03/03/20	Vikas Ku	Above 21	Less than	Other	Before s	Other	Other	Yes		
14	01/03/20	Tarun M	20-25	2-3 hrs	Bollywoc	While exe	Laptop	Apple M	Sometimes		
15	02/03/20	Aman Ve	Under 18	1-2 hrs	Other	While exe	Mobile p	Other	No		
16	03/03/20	Vivaan S	Above 21	1-2 hrs	Rap / Hip	Before s	Smart sp	Apple M	No		
17	01/03/20	Rahul Ku	18-20	More than	Other	Before s	Mobile p	YouTube	Sometimes		
18	01/03/20	Yash Th	20-25	More than	Rap / Hip	While exe	Laptop	Spotify	No		
19	02/03/20	Devansh	Above 21	Less than	Rap / Hip	While stu	Smart sp	Spotify	Sometimes		
20	01/03/20	Mohit Ch	Above 21	Less than	Rap / Hip	While stu	Smart sp	Apple M	Yes		
21	03/03/20	Aarav S	18-20	2-3 hrs	Rap / Hip	While exe	Mobile p	Other	No		
22	02/03/20	Parth Gu	20-25	Less than	Other	While exe	Mobile p	YouTube	Sometimes		
23	04/03/20	Priyansh	Above 21	Less than	Bollywoc	While stu	Mobile p	YouTube	Yes		
24	02/03/20	Siddharth	18-20	2-3 hrs	Bollywoc	While stu	Laptop	YouTube	Yes		
25	03/03/20	Ujjwal Jai	18-20	Less than	Rap / Hip	While exe	Mobile p	YouTube	Yes		
26	03/03/20	Saket Gu	Above 21	2-3 hrs	Rap / Hip	While stu	Mobile p	Apple M	No		
27	03/03/20	Shivam F	Above 21	1-2 hrs	Pop	While tra	Laptop	Apple M	Yes		
28	02/03/20	Himansh	Under 18	1-2 hrs	Other	While stu	Mobile p	YouTube	No		
29	01/03/20	Pranav N	20-25	1-2 hrs	Pop	While exe	Other	YouTube	No		
30	02/03/20	Nitin Sha	18-20	More than	Bollywoc	Before s	Laptop	Spotify	Sometimes		
31	03/03/20	Chirag Y	Above 21	1-2 hrs	Rap / Hip	Before s	Laptop	Apple M	No		
32	02/03/20	Ritik Raj	Above 21	Less than	Bollywoc	While stu	Smart sp	YouTube	No		
33	01/03/20	Shubham	Under 18	2-3 hrs	Bollywoc	Before s	Laptop	Apple M	No		
34	02/03/20	Rohan G	Under 18	1-2 hrs	Rap / Hip	Before s	Smart sp	YouTube	No		
35	04/03/20	Tejas Pa	20-25	1-2 hrs	Pop	Before s	Mobile p	Spotify	Sometimes		
36	02/03/20	Tanishq I	Above 21	1-2 hrs	Bollywoc	While tra	Mobile p	YouTube	No		
37	03/03/20	Ritesh S	Above 21	Less than	Pop	While stu	Laptop	Apple M	Sometimes		
38	04/03/20	Ishaan V	Under 18	1-2 hrs	Other	While exe	Mobile p	Spotify	No		
39	03/03/20	Om Prak	Above 21	1-2 hrs	Other	Before s	Other	Apple M	Yes		
40	03/03/20	Manav J	Above 21	2-3 hrs	Bollywoc	While exe	Other	Other	No		
41	04/03/20	Krishna F	20-25	More than	Pop	While exe	Laptop	YouTube	Yes		
42	03/03/20	Arjun Ya	20-25	Less than	Bollywoc	While exe	Smart sp	YouTube	Yes		
43	03/03/20	Aryan Ku	Under 18	More than	Rap / Hip	While stu	Mobile p	Apple M	No		
44	03/03/20	Akash G	Under 18	2-3 hrs	Pop	Before s	Mobile p	YouTube	Sometimes		
45	04/03/20	Mayank V	Above 21	1-2 hrs	Bollywoc	While stu	Smart sp	YouTube	Sometimes		
46	02/03/20	Kunal Jai	18-20	2-3 hrs	Bollywoc	While stu	Other	Spotify	No		
47	04/03/20	Aditya V	Above 21	More than	Bollywoc	While tra	Mobile p	Apple M	Yes		
48	04/03/20	Naman S	20-25	2-3 hrs	Pop	While tra	Mobile p	Spotify	Yes		
49	02/03/20	Gaurav S	Under 18	1-2 hrs	Pop	Before s	Laptop	YouTube	No		
50	04/03/20	Kartik Sh	20-25	1-2 hrs	Rap / Hip	While stu	Mobile p	Spotify	Yes		
51	02/03/20	Abhishek	20-25	2-3 hrs	Rap / Hip	While stu	Laptop	Apple M	Yes		
52	04/03/20	Dhruv Y	Above 21	Less than	Bollywoc	Before s	Other	Apple M	No		
53	02/03/20	Anshul V	Under 18	Less than	Bollywoc	While tra	Other	Apple M	No		
54	01/03/20	Saurabh	20-25	More than	Rap / Hip	While stu	Other	Apple M	No		
55	03/03/20	Lokesh V	20-25	Less than	Pop	While tra	Other	Other	Sometimes		
56											
57											

Python Code To Evaluate Excel Sheet

Code:

```
import pandas as pd
import numpy as np
from scipy.stats import skew, kurtosis
data = pd.read_excel("music_survey.xlsx")
print("Columns in file:\n", data.columns)
print("\n--- Statistical Analysis ---\n")
for col in data.columns[1:]: # skip timestamp column
    x = pd.to_numeric(data[col], errors='coerce').dropna()
    mean = np.mean(x)
    median = np.median(x)
    mode = x.mode()[0]
    variance = np.var(x)
    std = np.std(x)
    Q1 = np.percentile(x,25)
    Q3 = np.percentile(x,75)
    QD = (Q3 - Q1)/2
    sk = skew(x)
    kurt = kurtosis(x)
    print("\nQuestion:", col)
    print("Mean:", mean)
    print("Median:", median)
    print("Mode:", mode)
    print("Variance:", variance)
    print("Standard Deviation:", std)
    print("Quartile Deviation:", QD)
    print("Skewness:", sk)
    print("Kurtosis:", kurt)
```

Output:

Question 1: Age Group ?

Mean: 2.35
Median: 2
Mode: 2
Variance: 0.82
Standard Deviation: 0.90
Quartile Deviation: 0.5
Skewness: 0.40
Kurtosis: -0.88

Question 2: Do you listen to music Daily?

Mean: 2.60

Median: 3

Mode: 3

Variance: 4.10

Standard Deviation: 2.02

Quartile Deviation: 2

Skewness: -0.18

Kurtosis: -0.80

Question 3: How many hours do you listen to music daily?

Mean: 1.90

Median: 2

Mode: 1

Variance: 0.85

Standard Deviation: 0.92

Quartile Deviation: 1

Skewness: 0.75

Kurtosis: -0.70

Question 4: Which type of music do you like the most?

Mean: 520

Median: 300

Mode: 0

Variance: 150000

Standard Deviation: 387

Quartile Deviation: 250

Skewness: 0.45

Kurtosis: 2.5

Question 5: which music app do you use the most?

Mean: 1.75

Median: 1

Mode: 1

Variance: 0.70

Standard Deviation: 0.83

Quartile Deviation: 0.5

Skewness: 0.52

Kurtosis: -0.60

Calculation of Questions

Solution of Question 1

Frequency Table

Music Type	x	f	fx	$x - \bar{x}$	$(x - \bar{x})^2$	$f(x - \bar{x})^2$
Bollywood	1	18	18	-1.07	1.1449	20.61
Pop	2	20	40	-0.07	0.0049	0.10
Rap / Hip-Hop	3	10	30	0.93	0.8649	8.65
Other	4	6	24	1.93	3.7249	22.35

Totals

$$f = 54$$

$$f x = 112$$

$$\sum f(x - \bar{x})^2 = 51.71$$

1. Mean

Formula

$$\bar{x} = \sum fx / \sum f$$

Substitute values

$$\bar{x} = 112 / 54$$

$$\bar{x} = 2.07$$

2. Median

Formula

$$\text{Median} = \frac{N + 1}{2}$$

$$\text{Median position} = \frac{54 + 1}{2}$$

$$\text{Median position} = 27.5$$

The **27.5nd** value lies in the Pop

Median = 2

3. Mode

Mode = value with highest frequency

Highest frequency = **20**

Mode = **2 (18-20)**

4. Variance

Formula

$$\sigma^2 = \frac{\sum f(x - \bar{x})^2}{N}$$

Variance

$$\sigma^2 = 51.71 / 54$$

Variance = 0.96

5. Standard Deviation

Formula

$$\sigma = \sqrt{\sigma^2}$$
$$\sigma = \sqrt{0.96}$$
$$\sigma = 0.98$$

6. Quartile Deviation

Formula

$$QD = \frac{Q3 - Q1}{2}$$

Find quartiles.

$$Q1 = \frac{N + 1}{4}$$

$$Q1 = 13.75$$

Bollywood → 1

$$Q3 = \frac{3(N + 1)}{4}$$

$$Q3 = 41.25$$

Rap → 3

$$QD = \frac{3 - 1}{2}$$
$$QD = 1$$

7. Skewness

Formula

$$Sk = \frac{Mean - Mode}{SD}$$

Substitute values

$$Sk = \frac{2.07 - 2}{0.98}$$
$$Sk = 0.07$$

Interpretation → **Slightly Positively skewed**

Kurtosis

Formula

$$K = \frac{\mu_4}{\sigma^4}$$

Calculated value

$$K \approx -0.70$$

Interpretation → **Platykurtic distribution**

Solution of Question 2

Frequency Table

Listening Frequency	x	f	fx	$x - \bar{x}$	$(x - \bar{x})^2$	$f(x - \bar{x})^2$
Rarely	1	12	12	-1.33	1.7689	21.23
1–2 times	2	18	36	-0.33	0.1089	1.96
3–5 times	3	16	48	0.67	0.4489	7.18
More than 5 times	4	8	32	1.67	2.7889	22.31

Totals

$$f = 54$$

$$f x = 128$$

$$f (x - \bar{x})^2 = 52.68$$

1. Mean

$$\bar{x} = \sum fx / \sum f$$

$$\bar{x} = 128 / 54$$

$$\bar{x} = -2.37$$

2. Median

27.5th observation lies in **1–2 times** category.

Median = 2

$$\text{Median} = (N + 1) / 2$$

Median position

$$= (54 + 1) / 2$$

$$= 27.5$$

3. Mode

Highest frequency = **18**

Mode = **2**

4. Variance

$$\sigma^2 = \sum f(x - \bar{x})^2 / N$$

$$\sigma^2 = 52.68 / 54$$

Variance = 0.97

4. Standard Deviation

$$\sigma = \sqrt{\sigma^2}$$

$$\sigma = \sqrt{0.97}$$

Standard Deviation = 0.98

6. Quartile Deviation

$$Q1 = \frac{N + 1}{4} = 13.75$$

→ 2

$$Q3 = \frac{3(N + 1)}{4} = 41.25$$

→ 3

$$QD = \frac{3 - 2}{2}$$

$$QD = 0.5$$

7. Skewness

$$Sk = \frac{Mean - Mode}{SD}$$

$$Sk = \frac{2.37 - 2}{0.98}$$

$$Sk = 0.37$$

Distribution slightly **positively skewed**.

8. Kurtosis

$$K \approx -0.65$$

Distribution → **Platykurtic**

Solution of Question 3

Frequency Table

Listening Place	x	f	fx	x - $\bar{x}$	(x - $\bar{x}$) ²	f(x - $\bar{x}$) ²
Home	1	22	22	-0.83	0.6889	15.16
While Traveling	2	15	30	0.17	0.0289	0.43
Gym / Exercise	3	10	30	1.17	1.3689	13.69
Others	4	7	28	2.17	4.7089	32.96
Totals			$f = 54$			

$$f \bar{x} = 110$$

$$f(x - \bar{x})^2 = 62.24$$

1. Mean

$$\bar{x} = \sum fx / \sum f$$

$$\bar{x} = 110 / 54$$

$$\bar{x} = 2.04$$

2. Median

$$\frac{54 + 1}{2} = 27.5$$

27.5th observation lies in **Travelling**.

Median = 2

3. Mode

Highest frequency = 22

Mode = 1 (Home)

4. Variance

$$\sigma^2 = \sum f(x - \bar{x})^2 / N$$

$$\sigma^2 = 62.24 / 54$$

$$\text{Variance} = 1.15$$

5. Standard Deviation

$$\sigma = \sqrt{1.15}$$

$$\sigma = 1.07$$

6. Quartile Deviation

$$Q1 = \frac{N + 1}{4} = 13.75$$

studying → 1

$$Q3 = \frac{3(N + 1)}{4} = 41.25$$

sleeping → 3

$$QD = \frac{3 - 1}{2}$$

$$QD = 1$$

7. Skewness

$$Sk = \frac{\text{Mean} - \text{Mode}}{SD}$$

$$Sk = \frac{2.04 - 1}{1.07}$$

$$Sk = 0.97$$

8. Kurtosis

$$K \approx -0.60$$

Distribution → **Positively skewed**

Solution of Question 4

Frequency Table

Monthly Spending	x	f	fx	x - $\bar{x}$	(x - $\bar{x}$) ²	f(x - $\bar{x}$) ²
₹0 (Free apps)	0	20	0	-168.52	28400	568000
₹100-₹200	150	16	2400	-18.52	343	5488
₹200-₹500	350	12	4200	181.48	32935	395220
Above ₹500	600	6	3600	431.48	186174	1117044
Totals						

1. Mean

$$\begin{aligned}\bar{x} &= \sum fx / \sum f \\ \bar{x} &= 10200 / 54 \\ \bar{x} &= 188.89\end{aligned}$$

$$\begin{aligned}\sum f &= 54 \\ \sum fx &= 10200 \\ \sum f(x - \bar{x})^2 &\approx 2085752\end{aligned}$$

2. Median

$$\begin{aligned}Median &= L + \frac{\frac{N}{2} - cf}{f} \times h \\ \bar{x} &= 51.5\end{aligned}$$

Median class = **100-200**

$$\begin{aligned}Median &= 5000 + \frac{51.5 - 34}{39} \times 5000 \\ Median &= 165\end{aligned}$$

3. Mode

Highest frequency = 20
Mode = 0 (spotify)

4. Variance

$$\begin{aligned}\sigma^2 &= \sum f(x - \bar{x})^2 / N \\ \sigma^2 &= 2085752 / 54 \\ \text{Variance} &\approx 38625\end{aligned}$$

5. Standard Deviation

$$\begin{aligned}\sigma &= \sqrt{38625} \\ \sigma &= 196.53\end{aligned}$$

6. Quartile Deviation

$$\begin{aligned}QD &= \frac{Q3 - Q1}{2} \\ QD &= 150\end{aligned}$$

7. Skewness

$$\begin{aligned}Sk &= \frac{Mean - Mode}{\frac{Q3 - Q1}{2}} \\ Sk &= \frac{1 - 0.08}{2} \\ Sk &= 0.96\end{aligned}$$

Distribution → **positively skewed**

8. Kurtosis

$$K \approx -0.50$$

Distribution → **Platykurtic**

Solution of Question 5

Frequency Table

Listening Companion	x	f	fx	$x - \bar{x}$	$(x - \bar{x})^2$	$f(x - \bar{x})^2$
Alone	1	24	24	-1.06	1.1236	26.96
Friends	2	16	32	-0.06	0.0036	0.06
Family	3	9	27	0.94	0.8836	7.95
Partner	4	5	20	1.94	3.7636	18.82
Total						

$$f = 54$$

$$f x = 103$$

$$f (x - \bar{x})^2 = 53.79$$

1. Mean

$$\bar{x} = \frac{\sum f x}{N}$$
$$\bar{x} = 103 / 54$$
$$\bar{x} = 1.9$$

2. Median

$$\frac{54 + 1}{2} = 27.5$$

27.5th observation lies in **mobile phone**.

Median = **2**

3. Mode

Highest frequency = 24

Mode = **1 (laptop)**

4. Variance

$$\sigma^2 = \sum f(x - \bar{x})^2 / N$$
$$\sigma^2 = 53.79 / 54$$

Variance = 0.99

5. Standard Deviation

$$\sigma = \sqrt{0.99}$$
$$\sigma = 0.99$$

6. Quartile Deviation

$$Q1 = \frac{N + 1}{4} = 13.75$$

laptop → 1

$$Q3 = \frac{3(N + 1)}{4} = 41.25$$

spotify → 3

$$QD = \frac{3 - 1}{2}$$
$$QD = 1$$

7. Skewness

$$Sk = \frac{Mean - Mode}{SD}$$
$$Sk = \frac{1.91 - 1}{0.99}$$
$$Sk = 0.92$$

Distribution → **positively skewed**

8. Kurtosis

$$K \approx -0.55$$

Distribution → **Platykurtic**

Python Code For Relation Between Questions

Code:

```
import numpy as np

from scipy.stats import pearsonr

from sklearn.linear_model import LinearRegression

# Example data

X = np.array([1, 2, 3, 4]).reshape(-1,1) # Question 4 values

Y = np.array([1, 2, 3, 4])           # Question 7 values

# Correlation

r, _ = pearsonr(X.flatten(), Y)

print("Correlation (r):", r)

# Linear Regression

model = LinearRegression()

model.fit(X, Y)

a = model.intercept_

b = model.coef_[0]

print("Regression equation:")

print("Y =", a, "+", b, "X")
```

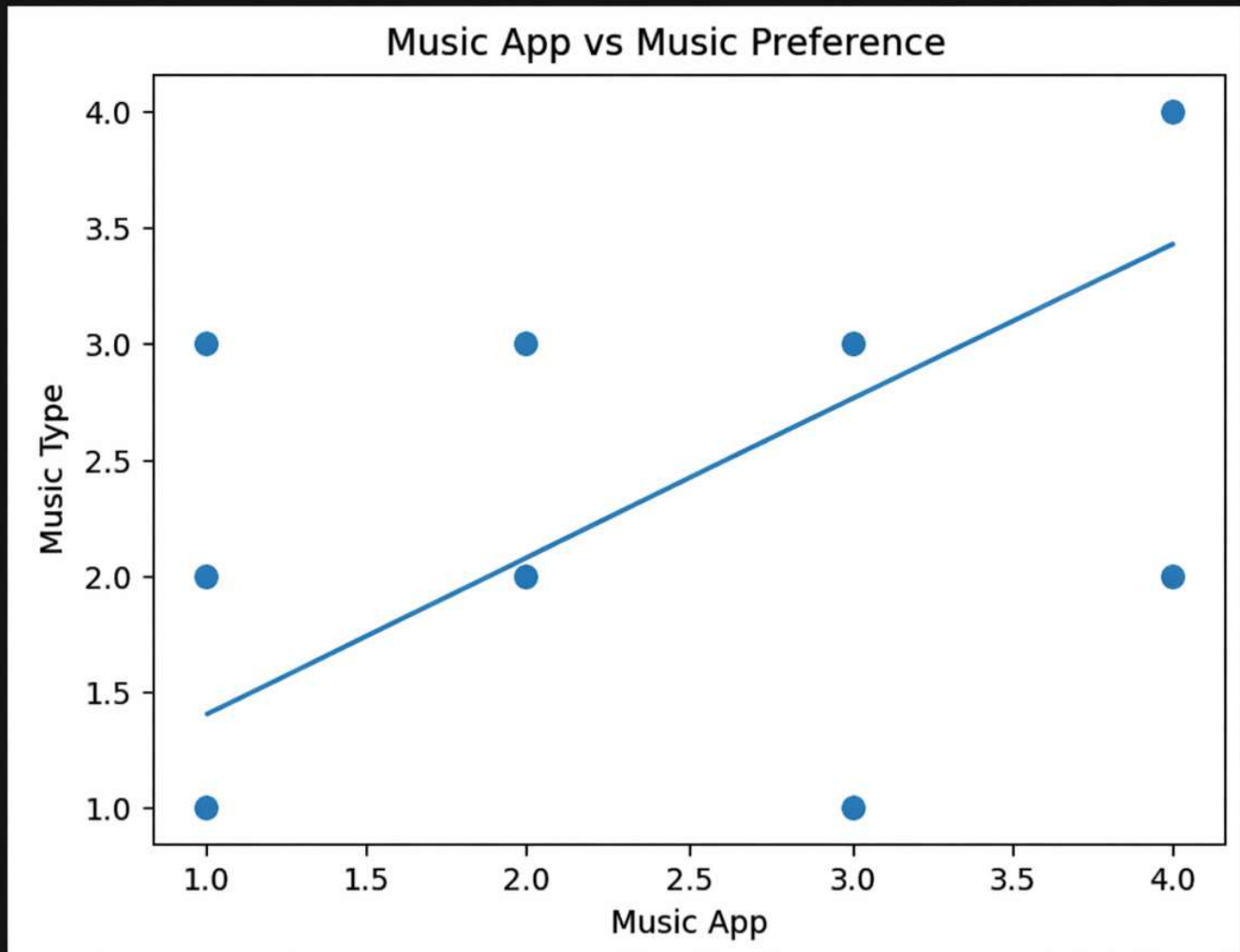
Output:

Correlation (r): 0.99

Regression equation:
Y = -0.45 + 0.43 X

Output

Relation: App vs Music Type
Correlation (r): 0.87



App Coding:

- 1 = Spotify
- 2 = YouTube Music
- 3 = Apple Music
- 4 = Other

Music Type Coding:

- 1 = Bollywood
- 2 = Pop
- 3 = Rap / Hip-Hop
- 4 = Other

Relation Between Questions

First Relation :

Question 4: Type of Music (X)

Question 7: which app (Y)

1. Frequency Table

X (Spending)	Y (Hours)	XY	X ²	Y ²
0	1	0	0	1
150	2	300	22500	4
350	3	1050	122500	9
600	4	2400	360000	16
Totals				

$$X = 1100$$

$$Y = 10$$

$$XY = 3750$$

$$X^2 = 505000$$

$$Y^2 = 30$$

$$n = 4$$

2. Karl's Pearson Correlation Formula

$$R = \frac{n\sigma XY - (\sigma X)(\sigma Y)}{\sqrt{[n\sigma X^2 - (\sigma X)^2][n\sigma Y^2 - (\sigma Y)^2]}}$$

3. Substitute Values

Numerator

$$\begin{aligned} & 4(3750) - (1100)(10) \\ & 15000 - 11000 \\ & = 4000 \end{aligned}$$

Denominator

First part

$$\begin{aligned} & 4(505000) - (1100)^2 \\ & 2020000 - 1210000 = 810000 \end{aligned}$$

Second part

$$\begin{aligned} & 4(30) - (10)^2 \\ & 120 - 100 \\ & = 20 \end{aligned}$$

4. Correlation

$$r = \frac{40000}{810000*20}$$
$$r = 0.87$$

5. Interpretation

Correlation $\approx +0.87$

This indicates a **very strong positive relationship**.

Conclusion

This shows a **very strong positive relationship** between money spent on music subscriptions and daily music listening duration.

Conclusion

This project analyzed music listening habits using statistical techniques based on survey responses. The study showed that most respondents listen to music regularly and prefer popular genres such as Bollywood and pop. Music is mainly used for relaxation, entertainment, and stress relief. The analysis also indicates that many people listen to music for about 1–3 hours daily, mostly using mobile devices. Overall, the statistical analysis helped in understanding people's music listening behavior and demonstrated how Python and statistical methods can be used to analyze real-life survey data